

## Technical Data Sheet

Date : 01.07.2025  
Product Name : Transformer Oil (Unused inhibited high grade insulating oil) - TRANSOL HGX  
Grade : IEC - 60296 (Edition 5.0 2020-06) - Inhibited, Type A (fully Inhibited high grade oils) & CLW Specification  
No. CLW/ES/3/0172  
Customer Name : Rev. No.01

| Sr. No.                                      | Property                                      | Test Method                                                                  | Limits                                                                                                                                                         | Typical Values                                              |
|----------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>1 - Function :</b>                        |                                               |                                                                              |                                                                                                                                                                |                                                             |
| 1                                            | Viscosity at 40 °C                            | ISO 3104                                                                     | Max. 12 mm <sup>2</sup> /s                                                                                                                                     | 10.8                                                        |
| 2                                            | Viscosity at - 30 °C                          | ISO 3104                                                                     | Max. 1800 mm <sup>2</sup> /s                                                                                                                                   | 1100                                                        |
| 3                                            | Pour Point                                    | ISO 3016                                                                     | Max. - 40 °C                                                                                                                                                   | - 42                                                        |
| 4                                            | Water Content                                 | IEC 60814                                                                    | Max. 30 mg / kg - for bulk supply                                                                                                                              | 12                                                          |
|                                              |                                               |                                                                              | Max. 40 mg /kg - delivery in drums/IB                                                                                                                          | 14                                                          |
| 5                                            | Breakdown voltage                             | IEC 60156                                                                    | Min. 30 kV as delivered                                                                                                                                        | 40 - 60                                                     |
|                                              |                                               |                                                                              | Min. 70 kV after laboratory treatment                                                                                                                          | > 72                                                        |
| 6                                            | Density at 20 °C                              | ISO 3675                                                                     | Max. 895 kg/m <sup>3</sup>                                                                                                                                     | 860                                                         |
|                                              | Density at 29.5 °C                            | ISO 3675                                                                     | Max. 890 kg/m <sup>3</sup>                                                                                                                                     | 855                                                         |
| 7                                            | DDF at 90 °C                                  | IEC 60247                                                                    | Max. 0.005                                                                                                                                                     | 0.00018                                                     |
| 8                                            | Aniline Point., °C                            | ASTM D 611                                                                   | <b>63 to 84 (RDSO/CLW SPEC.)</b>                                                                                                                               | 80                                                          |
| 9                                            | Carbon type composition                       | IEC 60590 & IS 13155 FTIR                                                    | <b>RDSO / CLW Spec.</b>                                                                                                                                        |                                                             |
|                                              | C <sub>A</sub>                                |                                                                              | Aromatics : 4 to 10 %                                                                                                                                          | 4.5                                                         |
|                                              | C <sub>P</sub>                                |                                                                              | Paraffins : < 50 %                                                                                                                                             | 48.5                                                        |
|                                              | C <sub>N</sub>                                |                                                                              | Naphthenic : > 42 %                                                                                                                                            | 47.0                                                        |
| <b>2 - Refining / Stability</b>              |                                               |                                                                              |                                                                                                                                                                |                                                             |
| 1                                            | Colour                                        | ISO 2049                                                                     | L 0.5 ( less than 0.5 )                                                                                                                                        | L 0.5                                                       |
| 2                                            | Appearance                                    | -                                                                            | Clear, free from sediment and suspended matter                                                                                                                 | Clear, free from sediment & suspended matter                |
| 3                                            | Acidity                                       | IEC 62021-2 or 62021-1                                                       | Max. 0.01 mg KOH/g                                                                                                                                             | 0.0014                                                      |
| 4                                            | Interfacial Tension                           | IEC 62961 or ASTM D 971                                                      | Min. 43 mN/m                                                                                                                                                   | 48                                                          |
| 5                                            | Total sulphur content                         | ISO 8754 / or ISO 20847                                                      | Max. 0.05 %                                                                                                                                                    | NIL                                                         |
| 6                                            | Corrosive Sulphur                             | DIN 51353                                                                    | Not corrosive                                                                                                                                                  | Non corrosive                                               |
|                                              | Potentially Corrosive Sulphur                 | IEC 62535                                                                    | Not corrosive                                                                                                                                                  | Non corrosive                                               |
| 7                                            | DBDS                                          | IEC 62697-1                                                                  | Not detectable ( < 5 mg/kg )                                                                                                                                   | Non detectable                                              |
| 8                                            | Inhibitors of IEC 60666                       | IEC 60666                                                                    | Inhibited Oil ( I ); 0.08 % to 0.4 %                                                                                                                           | 0.38                                                        |
| 9                                            | Metal passivator additives of IEC 60666       | IEC 60666                                                                    | Not detectable ( < 5 mg/kg ) or as agreed upon with the purchaser                                                                                              | Not detectable                                              |
| 10                                           | Other additives                               | -                                                                            | -                                                                                                                                                              | NIL (Not added)                                             |
| 11                                           | 2 - Furfural & related compounds content      | IEC 61198                                                                    | Not detectable ( < 0.05 mg/kg) for each individual compound                                                                                                    | Not detectable                                              |
| 12                                           | Stray gassing under thermo - oxidative stress | Procedure in Clause A.4 ( oil saturated with air ) in the presence of copper | Non stray gassing :<br>< 50 µl/l of hydrogen ( H <sub>2</sub> )<br>< 50 µl/l methane ( CH <sub>4</sub> )<br>< 50 µl/l ethane ( C <sub>2</sub> H <sub>6</sub> ) | Non stray gassing<br>Non stray gassing<br>Non stray gassing |
| <b>3 - Performance :</b>                     |                                               |                                                                              |                                                                                                                                                                |                                                             |
| 1                                            | Oxidation Stability at 120°C                  | IEC 61125 Test duration :<br>( I ) Inhibited Oil : 500 h                     |                                                                                                                                                                |                                                             |
| a)                                           | Total acidity                                 | 4.8.4 of IEC 61125: 2018                                                     | Max. 0.3 mg KOH / gm                                                                                                                                           | 0.14                                                        |
| b)                                           | Sludge                                        | 4.8.1 of IEC 61125: 2018                                                     | Max. 0.05 %                                                                                                                                                    | 0.012                                                       |
| c)                                           | DDF at 90 °C                                  | 4.8.5 of IEC 61125: 2018                                                     | Max. 0.05                                                                                                                                                      | 0.004                                                       |
| <b>4 - Health , safety and environment :</b> |                                               |                                                                              |                                                                                                                                                                |                                                             |
| 1                                            | Flash point                                   | ISO 2719                                                                     | Min. 140 °C                                                                                                                                                    | 148                                                         |
| 2                                            | PCA content                                   | IP 346                                                                       | < 3.0 %                                                                                                                                                        | 0.52                                                        |
| 3                                            | PCB content                                   | IEC 61619 / ASTM D 4059                                                      | Not detectable ( < 2 mg/kg )                                                                                                                                   | Not detectable                                              |

  
Authorized Signatory